

PART-BName of the Student: N. Rikhi ReddySignature of the Student: Rikhi Roll No.: 230123038Signature of the Invigilator: [Signature]**INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	NA	7	0	1	08

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{y\mathbf{X}}^T \\ S_{y\mathbf{X}} & S_{\mathbf{X}\mathbf{X}} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $\mathbf{X}_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector $\mathbf{X}$.

Q. 1. (a) Find the rank of the matrix

(b) Find the rank of the matrix

(c) Find the rank of the matrix

(d) Find the rank of the matrix

(e) Find the rank of the matrix

(f) Find the rank of the matrix

(g) Find the rank of the matrix

(h) Find the rank of the matrix

(i) Find the rank of the matrix

(j) Find the rank of the matrix

(k) Find the rank of the matrix

(l) Find the rank of the matrix

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
 - (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
 - (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

Given 4×4 sample covariance matrix $\mathbf{S}$
 eigen values are $\lambda_1 = 0.8, \lambda_2 = 2.1, \lambda_3 = 4.2, \lambda_4 = 0.2$. corresponding eigen vectors
 are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4$
 extract $m=2$ factors using principal component method

a) since $m=2$ we take the ~~eig~~ largest 2 eigen values and their
 corresponding eigen vectors for the estimated factor loading matrix $\tilde{\mathbf{L}}$

~~$\mathbf{L} = [\sqrt{\lambda_3} \mathbf{e}_3 | \sqrt{\lambda_2} \mathbf{e}_2]$~~ $\Rightarrow \lambda_2 = 2.1$ and $\lambda_3 = 4.2$ are used and their corresponding
 eigen vectors are $\mathbf{e}_2, \mathbf{e}_3$

explicit expression
 for $\tilde{\mathbf{L}}$ is

$$\tilde{\mathbf{L}} = [\sqrt{\lambda_3} \mathbf{e}_3 | \sqrt{\lambda_2} \mathbf{e}_2] \quad \leftarrow p \times 2 \text{ matrix with}$$

$\sqrt{\lambda_3} \mathbf{e}_3$ - first column

$\mathbf{e}_2, \mathbf{e}_3$ - $p \times 1$ matrices

$\sqrt{\lambda_2} \mathbf{e}_2$ - second column

b) given first row of $\tilde{\mathbf{L}} = [0.8, -0.4]$

estimated communality $\tilde{h}_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2$

$$= (0.8)^2 + (-0.4)^2$$

$$= 0.64 + 0.16$$

$$= 0.8$$

c) Given that the eigenvectors are normalised

$$\Rightarrow \sum_{k=1}^p e_{ik}^2 = 1 \quad \text{for } k=1, 2, 3, 4$$

Sum of all estimated communalities

$$\sum_{i=1}^p h_i^2 = \sum_{i=1}^p (l_{i1}^2 + l_{i2}^2)$$

$$= \sum_{i=1}^p (\sqrt{\lambda_3} e_{i3})^2 + (\sqrt{\lambda_2} e_{i2})^2$$

$$= \sum_{i=1}^p \lambda_3 e_{i3}^2 + \lambda_2 e_{i2}^2$$

$$= \lambda_3 \left(\sum_{i=1}^p e_{i3}^2 \right) + \lambda_2 \left(\sum_{i=1}^p e_{i2}^2 \right)$$

$$= \lambda_3 + \lambda_2 \quad \text{which is the sum of largest two eigenvalues of } S$$

hence sum of all estimated communalities $\sum_{i=1}^p h_i^2$ is exactly equal to the

sum of the largest $m=2$ eigenvalues of S .

04

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

let asymptotic variance of $\bar{X}_n = \sigma^2$

By delta method

$$\text{asymptotic variance of } g(\bar{X}_n) = [g'(\bar{X}_n)]^2 \cdot \sigma^2$$

also asymptotic variance of $X = \frac{1}{I_{\mu}(x)}$ Fisher information.

$\bar{X}_n = \frac{\sum X_i}{n}$, assuming X_i 's are independent normal random variables

$$\text{likelihood } L = \frac{1}{n} \cdot \frac{1}{(2\pi)^{n/2} \sigma^n} \cdot e^{-\frac{n(\bar{x}-\mu)^2}{2\sigma^2}}$$

$$\text{but given } \sigma^2 = \mu^2 \Rightarrow \sigma = \mu$$

$$L = \frac{1}{n} \cdot \frac{1}{(2\pi)^{n/2} \mu^n} \cdot e^{-\frac{n}{2} \left(\frac{\bar{x}-\mu}{\mu}\right)^2}$$

$$\begin{aligned} \log \text{likelihood } l &= -\log n - \frac{n}{2} \log 2\pi - n \log \mu - \frac{n}{2} \left(\frac{\bar{x}-\mu}{\mu}\right)^2 \\ &= -\log n - \frac{n}{2} \log 2\pi - n \log \mu - \frac{n\bar{x}^2}{2\mu^2} - \frac{n}{2} + \frac{n\bar{x}}{\mu} \end{aligned}$$

$$\frac{dl}{d\mu} = -\frac{n}{\mu} + \frac{n\bar{x}^2}{\mu^3} - \frac{n\bar{x}}{\mu^2}$$

$$\frac{d^2l}{d\mu^2} = -\frac{n}{\mu^2} + \frac{3n\bar{x}^2}{\mu^4} - \frac{2n\bar{x}}{\mu^3}$$

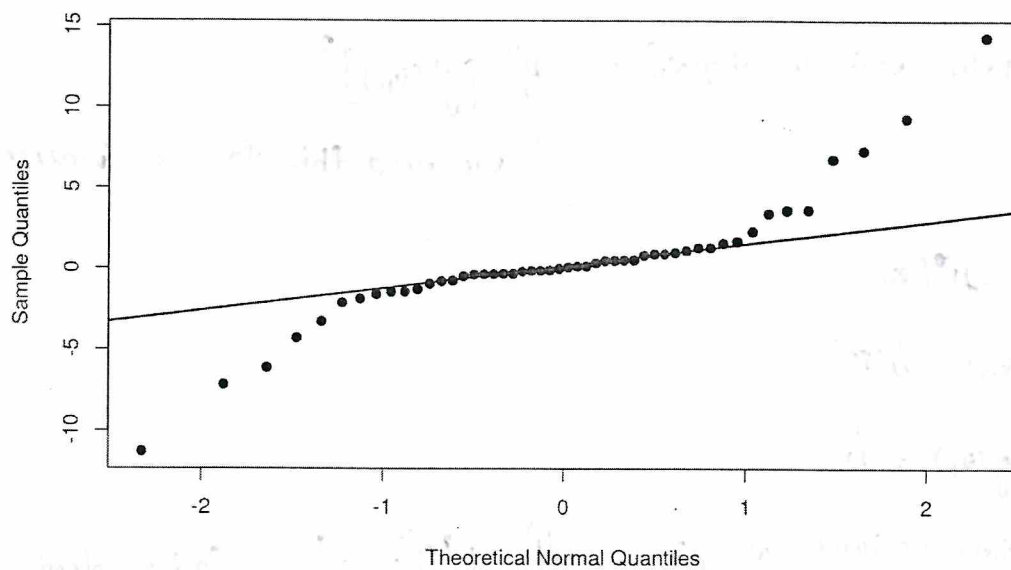
$$= \frac{n}{\mu^4} (-\mu^2 + 3\bar{x}^2 + 2\bar{x}\mu)$$

$$= \frac{n}{\mu^4} (3\bar{x}^2 - 2\bar{x}\mu - \mu^2)$$

$$= \frac{n}{\mu^4} (3(\bar{x}-\mu)^2 - 4\mu^2 + 4\bar{x}\mu)$$

$$\begin{aligned} I_{\mu}(\mu) &= E \left[-\frac{d^2l}{d\mu^2} \right] = E \left[\frac{n}{\mu^4} (3(\bar{x}-\mu)^2 - 4\mu^2 + 4\bar{x}\mu) \right] \\ &= \frac{n}{\mu^4} \left[3 E \left[\underbrace{(\bar{x}-\mu)^2}_{\text{var}(\bar{x})} \right] - 4\mu^2 + 4\mu E(\bar{x}) \right] \\ &= \frac{n}{\mu^4} [3 \cdot \mu^2 - 4\mu^2 + 4\mu^2] \\ &= 3n/\mu^2 \end{aligned}$$

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



We can observe from the Q-Q plot that the data has large deviation from normal for values >1 and <-1 which also shows similar factor of deviation to both positive and negative side, and close to point 2, -2 the values are around 8 which is close to suggesting that the data doesn't follow the normality assumption and exponential class of distribution will more accurately characterize the data.

01

3rd continued

asymptotic variance of $\bar{X}_n = \frac{\mu^2}{3n}$

$\Rightarrow$ asymptotic variance of $g(\bar{X}_n) = \frac{\mu^2}{3n} \cdot [g'(\bar{X}_n)]^2$
 we need this to be independent of μ

$\Rightarrow g(\bar{X}_n) = \frac{1}{\bar{X}_n}$

let $g(\bar{X}_n) = \frac{1}{\bar{X}_n}$

then $g'(\bar{X}_n) = -\frac{1}{\bar{X}_n^2}$

$\Rightarrow$ asymptotic variance of $g(\bar{X}_n) = \frac{\mu^2}{3n} \cdot \left(-\frac{1}{\mu^2}\right)^2 = \frac{1}{3n} \rightarrow$ independent of μ

So $g(\bar{X}_n) = \frac{1}{\bar{X}_n}$ is the transformation such that the asymptotic variance of $g(\bar{X}_n)$ is independent of μ .

